LabSentry AI

HACKATHON SUBMISSION | EDUCATIONAL PROTOTYPE

AI-Assisted Laboratory Specimen Pre-Check & Pre-Analytical Quality Dashboard Prototype

Architecture: Client-Side Edge Execution • 100% In-Browser
Workflow: Pre-Analytical Verification • Synthetic Demo Ledger

⚠ **Mandatory Educational Prototype Disclaimer:** LabSentry AI is an educational simulation prototype operating entirely on synthetic demo records. It is designed for workflow exploration and **does not diagnose patients, perform medical decision-making, or claim to prove that a specimen is clinically safe**. The system does not claim formal regulatory certification. Final specimen acceptance or redraw decisions remain solely with qualified laboratory professionals.

THE CLINICAL CONTEXT

Pre-Analytical Errors Lead to Sample Rejections and Delays

In clinical pathology, published literature highlights that pre-analytical steps (sample collection, labeling, tube selection, and handling) represent a major source of laboratory discrepancies and sample rejections before specimens reach automated analyzers. Typical workflow problems include collecting blood into incompatible additive tubes (e.g., EDTA instead of Citrate for coagulation assays), mismatched accession IDs between requisitions and vials, and underfilled tubes. In particular, underfilled Sodium Citrate tubes disrupt the necessary 9:1 blood-to-anticoagulant ratio, resulting in excess free citrate that can alter coagulation assay timings and necessitate specimen redraws.

THE PROPOSED SOLUTION

Automated Optical Scanning & Rule-Based Pre-Check Triage

LabSentry AI provides an interactive laboratory sample pre-check dashboard prototype. Running fully in the browser, the system combines client-side optical character recognition (OCR via Tesseract.js), image contrast pre-processing, deterministic clinical rule verification, and a human-in-the-loop review interface. The prototype cross-checks requisition IDs, sample IDs, tube additive types, and sample volume percentages in seconds, outputting transparent triage indicators—*No Mismatch Found*, *Review Required*, or *Mismatch Detected*—along with educational rationales for laboratory staff inspection.

SYSTEM WORKFLOW ARCHITECTURE

Sequential Ingestion & Verification Pipeline

1. Ingest	2. Vision & OCR	3. Review	4. Rules	5. Triage	6. Ledger
Order entry / label upload	Canvas boost + Tesseract	Technician verification	Tube & volume checks	Pass / Review / Alert	Audit log & CSV export

CORE IMPLEMENTED FEATURES (ALL 10 MODULES FUNCTIONAL)

Full Prototype Scope

1. Statistics Dashboard
Summary KPI cards: total sample checks, pass %, review %, discrepancy %, and volume compliance.

2. Clinical Intake Form
Order ID, Sample ID, Test Name, Expected Tube, Collected Tube, and 0–100% volume slider with vial visual.

3. Browser OCR Engine
Tesseract.js client OCR with canvas contrast stretching, grayscale binarization, and regex parsing.

4. Review & Edit Drawer
Side-by-side editing panel allowing technicians to correct misread fields before final validation.

5. Configured Rules Engine
Checks tube additive compatibility, accession matches, and tube-specific volume rules (e.g. 90% Citrate).

6. Clear Result Displays
Prominent outcome banners: **No Mismatch**, **Review Required**, or **Mismatch Detected**.

7. Clear Reason for Every Result
Categorized checklist with PASS/WARN/FAIL indicators and explanatory clinical rationales for every check.

8. Sample History & Filters
Searchable demo ledger with status and test filters, printable inspection slips, and CSV / JSON export.

9. Chart.js Analytics
Four live charts: Outcome Doughnut, Discrepancy Causes Bar, Dept Throughput, and Tube Color Mix.

10. Reset Demo Data & Studio
One-click reset to 12 synthetic demo cases plus Canvas barcode generator with 6 realistic test presets.

TECHNOLOGY STACK

Markup & Styling: HTML5, CSS3	`Tailwind`
Runtime Logic: Vanilla JavaScript	`ES6+ Modules`
Client-Side OCR: Tesseract.js	`v5.0 CDN`
Analytics Charts: Chart.js	`v4.4 CDN`
Data Persistence: Web Storage API	`LocalStorage`
Image Synthesis: HTML5 Canvas	`2D Context`

PROTOTYPE INNOVATIONS

Privacy-First Edge Architecture: All image parsing, OCR, and rules run strictly inside the user's browser without sending demo label images to external cloud endpoints.

Explainable Rationale Engine: Avoids opaque outputs by pairing every validation result with an explicit, educational rationale regarding additive compatibility and fill ratios.

Interactive Synthetic Test Studio: Built-in 2D Canvas barcode label generator with 6 curated scenarios for repeatable demonstration without external image dependencies.

LIMITATIONS & FUTURE SCOPE

Current Limitations: Evaluates simulated 2D label images and manual volume inputs; cannot measure physical clot presence or serum hemolysis, icterus, or lipemia (HIL indices).

Future Roadmap:

- Live WebRTC camera streaming for continuous barcode capture
- Computer vision liquid meniscus detection (TensorFlow.js)
- Integration with hospital LIMS standards (HL7 / FHIR)